Let $A \in \mathbb{R}^{n \times n}$ be a symmetric matrix.

Spectral decomposition theorem:

→ all eigenvalues of A are real.

$A = U \Lambda U^T$ where $U \in \mathbb{R}^{n \times n}$ is orthogonal matrix

and $\Lambda \in \begin{bmatrix} \lambda_1 & & & 0 \\ & \lambda_2 & & \\ & & \ddots & \\ 0 & & & \lambda_n \end{bmatrix} \in \mathbb{R}^{n \times n}$ where λ_i's are real eigenvalues of A.

$A = U \Lambda U^T \Rightarrow AU = U\Lambda \Rightarrow A \begin{bmatrix} | & & | \\ u_1 & \cdots & u_n \\ | & & | \end{bmatrix} = \begin{bmatrix} | & & | \\ u_1 & \cdots & u_n \\ | & & | \end{bmatrix} \begin{bmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_n \end{bmatrix}$

$\Rightarrow Au_i = \lambda_i u_i$ for $i = 1, 2, \cdots, n$ $\leftarrow (*)$

Particular subset of symmetric matrices.

→ positive definite / positive semi-definite matrices.

$S \in \mathbb{R}^{n \times n}$ (symmetric) is called positive definite if $\forall x \neq 0$, $x^T S x > 0$

semi definite $x^T S x \geq 0$

Q: Show that all eigenvalues of S are positive if S is positive definite?

Take $x = u_1$ in particular where u_1 is an eigenvector of S. $(S u_1 = \lambda_1 u_1)$

$$0 \leq u_1^T S u_1 = u_1^T \lambda_1 u_1 = \lambda_1 u_1^T u_1 \Rightarrow \lambda_1 > 0$$

Similarly, for positive semi-definite matrix, all eigenvalue are non-negative

Let A, B be any two matrices.

$A \in \mathbb{R}^{m \times n}$, $B \in \mathbb{R}^{n \times m}$

$AB \in \mathbb{R}^{m \times m}$ $\quad$ $BA \in \mathbb{R}^{n \times n}$

Non-zero eigenvalues of AB and BA are same.

Let $x \in \mathbb{R}^m$ be eigenvector of AB corresponding to a non-zero eigenvalue λ of AB.

$ABx = \lambda x$ $\Leftarrow$ given to us.

$\lambda Bx = B(\lambda x) = B(ABx) = (BA)(Bx)$

Bx is nonzero (why??)

If A & B are square invertible.

$B^{-1}BAB = AB$

$B^{-1}(BA)B$

$= AB$

Induced norm of a matrix.

(We have assumed the norm of vector)

1-D array $\rightarrow x \in \mathbb{R}^n$, $x^T x = \|x\|_2^2$, $\|x\|_2 = \sqrt{x^T x}$

- matrix as arrangement of numbers

Frobenius norm of A

$A = [a_{ij}]_{\substack{1 \le i \le n \\ 1 \le j \le n}}$

$$\|A\|_F = \sqrt{\sum_{j=1}^{n} \sum_{i=1}^{n} a_{ij}^2}$$

$$= \sqrt{\text{trace}(A^T A)}$$

$$A = \begin{bmatrix} a_{11} & \cdots & a_{1n} \\ a_{21} & \cdots & a_{2n} \\ \vdots & & \end{bmatrix}$$

$$\left.\begin{bmatrix} * \\ \vdots \\ * \end{bmatrix}\right\} \le n^2 \text{ - entries}$$

vectorization

$$\begin{bmatrix} * & * & * \\ \vdots & \vdots & \vdots \\ * & \cdots & * \end{bmatrix}_{n \times n}$$

2-D array

$A \in \mathbb{R}^{m \times n}$, $A : \mathbb{R}^n \to \mathbb{R}^m$ ← linear transformation.

norm $\|\cdot\|_2$ (on $\mathbb{R}^n$)

norm $\|\cdot\|_2$ (on $\mathbb{R}^m$)

$\forall x \in \mathbb{R}^n$, $x \longmapsto Ax$

$\forall x \neq 0$, $\max_{x \neq 0} \frac{\|Ax\|_2}{\|x\|_2} = \text{maxmag}(A) = \|A\|_2$

$\text{minmag}(A) = \min_{x \neq 0} \frac{\|Ax\|_2}{\|x\|_2} \geq 0$

Definition of $\|\cdot\|_2$ of matrix. $A: \mathbb{R}^n \longrightarrow \mathbb{R}^n$ (any norm on $\mathbb{R}^n$; any norm on $\mathbb{R}^n$)

A f^{n} $\|\cdot\|_2 : \mathbb{R}^{n\times n} \longrightarrow \mathbb{R}$ is called as norm if

i) $\|A\|_2 \geq 0 \quad \forall \ A \in \mathbb{R}^{n\times n}$ & (Positivity of norm)

$\|A\|_2 = 0$ if & only if $A = 0$

ii) $\|\alpha A\|_2 = |\alpha| \|A\|_2 \quad \forall \ \alpha \in \mathbb{R}, \ \neq 0, \ A \in \mathbb{R}^{n\times n}$ (Absolute homogenity)

iii) $\|A+B\|_2 \leq \|A\|_2 + \|B\|_2 \quad \forall \ A, B \in \mathbb{R}^{n\times n}$ (triangle inequility)

iv) $\|AB\|_2 \leq \|A\|_2 \ \|B\|_2$ (Submultiplicativity)

We want to show that $\|\cdot\|_2 : \mathbb{R}^{n\times n} \to \mathbb{R}$ is a norm.

defined as

$$\|A\|_2 = \max_{x\neq 0} \frac{\|Ax\|_2}{\|x\|_2}$$

(Think of $\|\cdot\|_2$ as g)

$$g(A) = \max_{x\neq 0} \frac{\|Ax\|_2}{\|x\|_2}$$

i) $\|A\|_2 \geq 0 \quad \forall A \in \mathbb{R}^{n\times n}$

$$\|A\|_2 = \max_{x\neq 0} \frac{\|Ax\|_2}{\|x\|_2} \geq 0$$

Q: why $\|A\|_2$ exist (finite)

$\frac{\|Ax\|_2}{\|x\|_2}$ is bounded above.

$\max \|Ay\|_2$
$\|y\|_2 = 1$

SVD Theorem:

Let $A \in \mathbb{R}^{m \times n}$ and $\operatorname{rank}(A) = r$.

Then $\exists$ orthogonal matrices $U \in \mathbb{R}^{m \times m}$ and $V \in \mathbb{R}^{n \times n}$ and $\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_r > 0$ such that $\Sigma = \left[\begin{array}{c|c} \begin{matrix} \sigma_1 & & 0 \\ & \sigma_2 & \\ 0 & & \sigma_r \end{matrix} & 0 \\ \hline 0 & 0 \end{array}\right] \in \mathbb{R}^{m \times n}$ such that

$$A = U \Sigma V^T$$

$$U = \begin{bmatrix} | & | & & | \\ u_1 & u_2 & \cdots & u_m \\ | & | & & | \end{bmatrix} \in \mathbb{R}^{m \times m} \qquad U^T U = U U^T = I_m$$

$$V = \begin{bmatrix} | & | & & | \\ v_1 & v_2 & \cdots & v_n \\ | & | & & | \end{bmatrix} \in \mathbb{R}^{n \times n} \qquad V^T V = V V^T = I_n$$

$$\Sigma = \left[\begin{array}{ccc|c} \sigma_1 & & & \\ & \sigma_2 & & \\ & & \ddots & \\ & & \sigma_r & \\ \hline & 0 & & 0 \end{array}\right] \in \mathbb{R}^{m \times n}$$

$$A = U \Sigma V^T = \underset{m \times m}{\begin{bmatrix} | & | & & | \\ u_1 & u_2 & \cdots & u_m \\ | & | & & | \end{bmatrix}} \left[\begin{array}{ccc|c} \sigma_1 & & & \\ & \ddots & & \\ & & \sigma_r & \\ \hline & 0 & & 0 \end{array}\right]_{m \times n} \underset{n \times n}{\begin{bmatrix} - & v_1^T & - \\ - & v_2^T & - \\ & \vdots & \\ - & v_n^T & - \end{bmatrix}}$$

$$\Rightarrow AV = U\Sigma \Rightarrow A \begin{bmatrix} | & | & & | \\ v_1 & v_2 & \cdots & v_n \\ | & | & & | \end{bmatrix} = \begin{bmatrix} | & | & & | \\ u_1 & u_2 & \cdots & u_m \\ | & | & & | \end{bmatrix} \left[\begin{array}{ccc|c} \sigma_1 & & & \\ & \ddots & & \\ & & \sigma_r & \\ \hline & 0 & & 0 \end{array}\right]$$

$$\Rightarrow \quad A v_1 = \sigma_1 u_1$$
$$\vdots$$
$$A v_r = \sigma_r u_r$$
$$A v_{r+1} = 0$$
$$\vdots$$
$$A v_n = 0$$

$$N(A) = \text{span}\{v_{r+1}, \dots, v_n\}$$
$$\| \quad \text{Im}(A) = \text{span}\{u_1, u_2, \dots, u_r\}$$
$$\text{Ker}(A)$$

$$A \in \mathbb{R}^{m \times n}$$

$$\Rightarrow A : \mathbb{R}^n \longrightarrow \mathbb{R}^m$$

$$\text{Ker}(A) = N(A) = \{x \in \mathbb{R}^n \mid Ax = 0\}$$

$$A = U \Sigma V^T$$
$$= A^T = V \Sigma^T U^T$$
$$= A^T U = V \Sigma^T$$
$$\Rightarrow A^T u_1 = \sigma_1 v_1$$
$$A^T u_2 = \sigma_2 v_2$$
$$\vdots$$
$$A^T u_r = \sigma_r v_r$$
$$A^T u_{r+1} = 0$$
$$\vdots$$
$$A^T u_m = 0$$

$$N(A^T) = \text{span}\{u_{r+1}, \dots, u_m\}$$
$$\text{Im}(A^T) = R(A^T) = \text{span}\{v_1, v_2 \dots, v_r\}$$

$\{v_1, v_2 \dots, v_n\}$ is orthonormal basis of $\mathbb{R}^n$.

$$A = \underset{m\times m}{\begin{bmatrix} | & | & & | \\ u_1 & u_2 & \cdots & u_m \\ | & | & & | \end{bmatrix}} \underset{m\times n}{\begin{bmatrix} \sigma_1 & & & 0 \\ & \ddots & & \\ & & \sigma_r & \\ 0 & & & 0 \end{bmatrix}} \underset{n\times n}{\begin{bmatrix} - & v_1^T & - \\ - & v_2^T & - \\ & \vdots & \\ - & v_n^T & - \end{bmatrix}}$$

$\tilde{U}\tilde{\Sigma}\tilde{V}^T$

$$A = \underset{m\times r}{\begin{bmatrix} | & | & & | \\ u_1 & u_2 & \cdots & u_r \\ | & | & & | \end{bmatrix}} \underset{r\times r}{\begin{bmatrix} \sigma_1 & & 0 \\ & \ddots & \\ 0 & & \sigma_r \end{bmatrix}} \underset{r\times n}{\begin{bmatrix} - & v_1^T & - \\ - & v_2^T & - \\ & \vdots & \\ - & v_r^T & - \end{bmatrix}}$$

← Trimmed SVD)

$$= \underset{m\times r}{\begin{bmatrix} | & | & & | \\ \sigma_1 u_1 & \sigma_2 u_2 & \cdots & \sigma_r u_r \\ | & | & & | \end{bmatrix}} \underset{r\times n}{\begin{bmatrix} - & v_1^T & - \\ - & v_2^T & - \\ & \vdots & \\ - & v_r^T & - \end{bmatrix}}$$

$$\boxed{A = \sigma_1 u_1 v_1^T + \sigma_2 u_2 v_2^T + \cdots + \sigma_r u_r v_r^T} \leftarrow$$

for any $a \in \mathbb{R}^{m\times 1}$ & $b \in \mathbb{R}^{n\times 1}$, $ab^T = \begin{bmatrix} a_1 \\ \vdots \\ a_m \end{bmatrix} \begin{bmatrix} b_1 & \cdots & b_n \end{bmatrix}$

$$= \begin{bmatrix} a_1b_1 & \cdots & a_1b_n \\ a_2b_1 & \cdots & a_2b_n \\ \vdots & & \\ a_mb_1 & \cdots & a_mb_n \end{bmatrix}_{m\times n}$$

$$= \begin{bmatrix} b_1 \begin{vmatrix} \\ a \\ \end{vmatrix} & b_2 \begin{vmatrix} \\ a \\ \end{vmatrix} & \cdots & b_n \begin{vmatrix} \\ a \\ \end{vmatrix} \end{bmatrix}$$

$$A = U \Sigma V^T$$

$$A^T A = V \Sigma^T U^T U \Sigma V^T = V \Sigma^T \Sigma V^T$$

$$A A^T = U \Sigma V^T V \Sigma^T U^T = U \Sigma \Sigma^T U^T$$

Q: If A is a positive definite matrix, then what is its SVD ??

Q: If A is symmetric with some -ve eigenvalues, then what is its SVD ??

Q: Compute SVD of $\begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$